



# A three-dimensional software framework for environmental system monitoring and decision support in Poyang lake basin

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## Abstract

With the development of remote sensing and large-scale environmental modelling, large amount of environmental data are continuously becoming available. An intuitive and comprehensive visualization of these data could facilitate data exploration, communication and collaboration between the stakeholders for informed decisions making. In Poyang lake basin regions, we demonstrate how to develop a software platform that can visualize three-dimensionally environmental data layers including terrain, weather, river net, water level, land use changes and interrelations between these data layers for environmental system monitoring and decision supporting. The tool is built by combining several prevailing projects including Unity, Paraview, and hydrological models, etc. We develop an open standardized framework for the software tool to host environmental data layers permitting the application of the tool, which could be employed for the intuitive and comprehensive data visualization in other regions facing environmental challenges.

**Keywords** Environmental information system · Intuitive visualization · Poyang lake basin · Unity 3D · Data layer integration

## Introduction

With the development of remote sensing and large scale environmental modelling, many key environmental data are continuously and increasingly available for public access and supporting policy making (Guo et al. 2014). These data are valuable for relevant stakeholders and decision-makers (Wang and Yuan 2014). However, these data on environmental issues are often spatiotemporal, multivariate, stem from different sources, and are very often generated from multiple simulations run from numerous simulations models (Kumpf et al. 2018). To utilize these data, exploration of these data for collaboration between scientists from different fields are

indispensable (Rink et al. 2014a). Two issues hinder the exploitation of these data and discount their potential value. On one hand, data from different sources are frequently of high complexity and heterogeneity, and often in a variety of formats such as reports, raw imageries, or csv files which are not easily interpretable. On the other side, the vertical dimension of these data layers when presented are very often ignored, while it is of great importance for the horizontal flow of key elements (e.g. water) in an environment system. An intuitive visualization in three dimension could be very helpful to explore these data and get the insight into them (Wijk 2005), thereby enabling resource managers make informed decisions. Furthermore, it can promote interdisciplinary communication, create opportunities for more intuitive applications of environmental research, and convey the complex environmental processes to a wider audience (Grainger et al. 2016; Rink et al. 2014b).

Environmental data visualization is playing a key role in many fields including water resource management, and an increasing number of environmental data visualization systems are built for a variety of environmental applications (Grainger et al. 2016; Rink et al. 2014b). For instance, a framework is developed for exploring, analysing and comparing spatial-temporal climate data (Janicke et al. 2009). A web based visualization tool is presented for large climate data sets

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display (Alder and Hostetler 2015), which can achieve high visualization efficiency, and process the large scale data with the data size reaching tens of terabytes level. An efficient visualization solution for climate simulation data is presented (Cao et al. 2016), which can provide the scientists a deeper understanding of the multi-field climate data. Additionally, a workflow is developed for 3D visualization of atmospheric data in a virtual reality environment (Helbig et al. 2014). A multi-view approach is developed for visual analytics of smog in China (Li et al. 2016), which enables the identification of smog sources. A web based mapping visualization system is developed for urban air quality monitoring data (Lu et al. 2017), which achieves an intuitive visualization and enables significant pollution findings at the nationwide scale. Based on Google earth, the environmental wind flow and pollution dispersion are visualized (Liu and Kenjeres 2017), which present the relevant users a photo-realistic result. A spatiotemporal visualization solution is proposed for marine environmental data on South China Sea (He et al. 2010). A software tool is developed for interactive visualization of hydrodynamics flux for the coastal zone (George et al. 2014), which assists the ocean scientists for their research.

Furthermore, to realize high immersion of visualization technically, game engines especially unity is increasingly introduced to environmental data visualization (Broderick et al. 2016b; Gang et al. 2016; Helbig et al. 2014; Manyoky et al. 2014; Rafiee et al. 2017). For marine research, a unity based system is presented for marine visualization (Broderick et al. 2016b), which allows users visually and collaboratively discuss datasets and make informed decisions. Additionally, large scale marine hydrological environmental data is visualized by Su et al. (2016), which meets the requirements for the immersive ocean data visualization and real-time interaction of ocean scientists. For flood, a unity based visualization system for hydrological and water hazard information is developed by Gang et al. (2016), which enables real-time observation and disaster prediction by combining with 3D GIS. Based on unity and real life GIS data, a storm surge flooding animation system is presented, which can provide the users with immersive experiences (Reyes et al. 2017). For environmental monitor and evaluation, an integrated 3D visualization system is developed to assist resource managers and regulators to assess, monitor and make informed decisions on ground water management (Hunter et al. 2016). A game engine based 3D visualization of wind parks is proposed for public impact assessment of wind farms (Manyoky et al. 2014). For environmental planning and siting, based on unity, an interactive information system is proposed for wind turbine siting, which integrates game engine, GIS and analytical sound propagation model in a unified 3D web environment (Rafiee et al. 2017), thereby enabling the users an interactive, immersive and real-world siting tool. Additionally, based on unity, an urban heat island visual analysis solution is developed, which provides

the users with urban heat explorations and analysis (Klein 2016). An interactive visual interface that enables simultaneous visualization of ensemble weather forecast analyses data is proposed (Kumpf et al. 2018), guiding the meteorologists towards cluster data robustness information required for the weather analysis.

Although numerous visualization methods are developed, there are two issues which still remain to be addressed. Firstly, while previous researches developed many methods of interactive visualization of spatial-temporal environmental data for data exploration and decision support in the scope of specific domain, there is still lack of generalized method that can be easy to be used in other domains (Bilke et al. 2014; Chen et al. 2015). Faced with the high complexity of the environmental data visualization problems, it is rather difficult to build an environmental system that can visualize relevant datasets in 3D immersive environment and provide an intuitive, clear and direct understanding of current environment status from scratch. Secondly, while 3D game engines can present users a vivid and visually attractive scenes (Broderick et al. 2016a), they favour real-time rendering over data accuracy and physical correctness (Frieze et al. 2008). Fortunately, besides the visually attractive scenes, the game engines also can provide excellent modular functions for rendering, visualization, and interactions. Furthermore, they can make the basic setup of creation of 3D world ready and manage numerous data involved in the game, which significantly reduce the complexity of visualization system building. While these features indicate that they have potential for use in non-entertainment or real-world projects (Broderick et al. 2016a), a common strategy to link these features with real world data should be established, which requires that the game engine should be adapted to the very high accuracy and physical correctness of scientific datasets.

In this study, a software tool utilizing unity is presented for 3D integrative visualization of environmental data of Poyang lake basin involving large scale spatiotemporal heterogeneous datasets, which is characterized as high scalability, interactivity, and immersion in a virtual geographic environment. A common strategy to three-dimensionally visualize geospatial environmental data employing game engines is proposed.

## Materials and methodology

### Research area

Poyang lake basin is located in southeast China, spanning from the bank of the middle lower reaches of Yangtze River in the north into hillier areas in the south and east. With an area of 162,000 km<sup>2</sup>, representing 97.2% of Jiangxi province, the boundary of Poyang lake basin coincides very well with the

administrative boundary of Jiangxi province. Mountains surround Poyang lake basin on three sides, with the Mufu Mountains, Jiuling Mountains, and Luoxiao Mountains on the west; Huaiyu Mountains and Wuyi Mountains on the east; and the Jiuling Mountains and Dayu Mountains on the south. The elevation of the basin ranges from just a few meters in the lower reaches of the primary watercourses to about 2,200 metres in highly mountainous regions, yielding a heterogeneous topography of Poyang lake basin. Poyang Lake situates in the north of the basin. It is the largest freshwater lake in China (Du et al. 2018). It is fed by the Gan, Fu, Xin, Xiu, Rao rivers from West, South and East, and connected to the Yangtze through a channel. These tributaries cover the whole basin. The lake is a favourite destination for birding that provides a habitat for half a million migratory birds. During the winter, the lake becomes home to a large number of migrating Siberian cranes, up to 90% of which spend the winter there. Poyang Lake plays a significant role on the health and security of Poyang lake basin.

With the rapid development of local economy around the lake in the past decades, the water resources are over exploited, which lead to environmental problems, such as water quality deterioration and lake area shrinkage. Lake area shrinkage also leads to reduced biodiversity (Bing et al. 2018; Soldatova et al. 2018). These problems put severe threats to the local ecological systems and the wellbeing of local habitats. A comprehensive environmental information system is in demand to have an overview of these issues for decision makers.

The required 3-dimensional environmental information system should have the capacity for integrating the available data sets, carrying out the efficient data exploration, figuring out the interrelations between the data sets, therefore promoting the communication between the relevant stakeholders and supporting sound decisions. An environmental visual integration framework will be introduced for data visualization and integration and an example will be demonstrated. The environmental information system will employ digital elevation model as the base layer, and is open for integration of a variety of relevant data layers. The current data layers to be integrated with relevant information are listed in Table 1. The preparation of these data layers will be detailed in the following section.

## Environmental data integration and visualization framework

As aforementioned, due to the high heterogeneity of environmental data, it is rather challenging to integrate the relevant data sets in a uniform context. To ease the heterogeneous data sets integration, a framework is developed to build the Poyang lake basin environmental information system. The steps are

described as follows. Firstly, a mesh representing the ground surface with triangular surface needs to be prepared. To generate the mesh, the size of the triangles for the mesh is established as the first step, and then the area surrounded by the Poyang Lake Basin border is triangulated, which yields a planar surface. Finally all the vertexes of the triangles should be mapped by the DEM model to formulate the final mesh. The mesh acts as the carrier of all the earth information and should be mapped by the real-world digital elevation model information. Secondly, the high resolution texture imaginaries are mapped to the generated mesh for the realistic real-world presentations. The imaginaries for texture surface are converted to the same GCS\_WGS\_1984 coordinate system in this study. Thirdly, data including spatial and temporal data is exhibited on the virtual environment. Simulation results that are generated by relevant environmental models, such as hydrological model, surface model, etc., are processed and prepared. As aforementioned, all the data is geo-referenced under the same coordinate system. Finally, once all the data get prepared, they are converted into a form supported by Unity and imported into Unity, utilizing the sophisticated functions of the game engine to build the immersive virtual world.

It should be noticed that a large scale texture or mesh should be split into tiles, due to the size limit that relevant software can process. The details of the framework are shown in Fig. 1.

## Models and software

The environmental data to integrate as data layers as digital elevation model (DEM), precipitation, temperature, and ecosystem surfaces should be first generated by some model. High accuracy surface modeling (HASM) is employed to obtain the enough precision due to its' advantage on generating high accuracy surfaces from sampling points (Yue 2011). Additionally, a depth-averaged two-dimensional hydrological model (Du et al. 2018), is applied to simulate the water level daily change of the Poyang lake. A land use model (Chen et al. 2014) is employed to simulate the land use change of Poyang lake from year 2000 to 2012.

Several software tools are involved. The data geo-referencing operation, projection definition, raster projection and mosaic of the spatial data layers are conducted inside ArcGIS (Bearman and Fisher 2012). OpenGeoSys Data Explorer is used for geometrical data processing and rendering (Rink et al. 2013), which is developed by the Helmholtz Centre for Environmental Research in Germany, providing the functionalities for creating mesh from the boundary data, mapping the mesh with the DEM, texturing the mesh and exporting to .vtk (<https://vtk.org/>) files supported by many 3d visualization software. ParaView, a software for realistic 3D visualization of scientific data, is used for texturing the mesh, and converting the .vtp (<https://vtk.org/>) files of the

**Table 1** The data information used in the Poyang Lake environmental information system.

| Data                                      | Format                 | Resolution | Coverage          | Source                                                                        |
|-------------------------------------------|------------------------|------------|-------------------|-------------------------------------------------------------------------------|
| Boundary data                             | Vector, esri shapefile | 30 m       | Poyang Lake basin | IGSNRR                                                                        |
| Boundary data                             | Vector, esri shapefile | 30 m       | Poyang Lake       | IGSNRR                                                                        |
| River net                                 | Vector, esri shapefile | 30 m       | Poyang Lake basin | UFZ                                                                           |
| DEM                                       | Raster, asc            | 90 m       | Poyang Lake basin | <a href="https://earthexplorer.usgs.gov/">https://earthexplorer.usgs.gov/</a> |
| Imaginary data                            | Raster, .tiff          | 30 m       | Poyang Lake basin | Google earth                                                                  |
| Land use data                             | Vector, esri shapefile | 30 m       | Poyang Lake basin | IGSNRR                                                                        |
| Climate-vegetation system                 | Raster, asc            | 2 km       | Poyang Lake basin | IGSNRR                                                                        |
| temperature, precipitation and land cover | Raster, asc            | 2 km       | Poyang Lake basin | IGSNRR                                                                        |
| Water level annual change                 | Raster, .tiff          | 2 km       | Poyang Lake       | IGSNRR                                                                        |
| Water level daily change data             | raster, .tiff          | 2 km       | Poyang lake       | IWHR                                                                          |

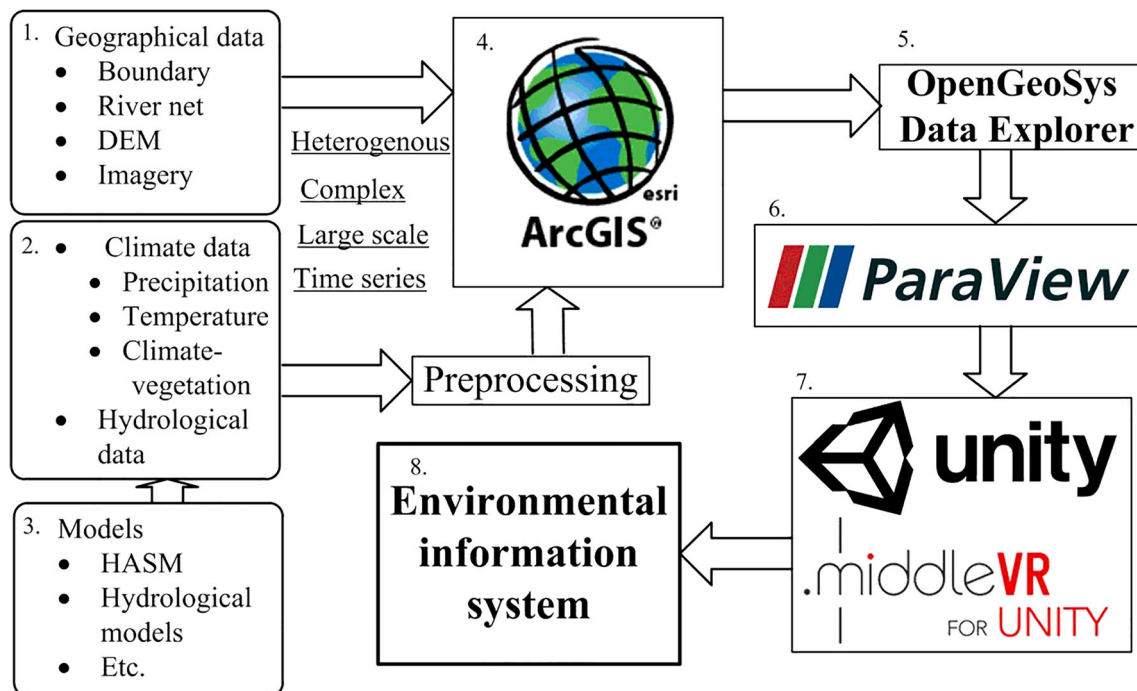
mesh to .fbx (<https://www.autodesk.com/products/fbx/overview>) files supported by unity with the developed plugin (Bilke et al. 2014).

Unity 3D, a popular game engine for 3D game development and virtual reality (Hocking 2015), is employed for intuitive visualization of the data layers. It allows users to develop games that can be published on multiple platforms such as PC, web, iOS, etc. Additionally, Unity is distributed under two available licenses, free and professional. The former shows a splash screen each time when your distributed game starts as well as lacks features for advanced features, while the latter enables the advanced features and has no splash screen.

It is worth noting that unity has an extremely active user community and the documents contains a wealth of valuable information (Blackman 2013), which makes Unity attractive to the users of the world.

MiddleVR, a virtual reality (VR) software develop kit (Kuntz 2015), which addresses the issues of all the aspects for creating a VR applications such as the input devices management, the interactions between the user and the world, etc., is employed for the interactions between the stakeholders and our visualization system.

Combined with MiddleVR, Unity can be used to design 3D game supported by Virtual Reality devices and support the



**Fig. 1** The framework of the EIS building. The parts are labeled from 1 to 8, where 1 and 2 represent the two kind heterogeneous data sets,

observation data and simulation data from models of 3, and 4, 5, 6 and 7 represent data processing tools employed



full interaction of the users with the 3D world. Therefore, it is adopted for our system development.

Elshayal Smart Web on Line Software (from <https://elshayal-smart.en.lo4d.com/>), a comprehensive and efficient software tool for adding, viewing and editing multiple map image files, is applied to download and geo-inference the satellite of the Google earth by screenshots the area of interests of Poyang lake basin.

### Data preparation

Here we describe the steps for the preparations of primary spatial data layers (Table 1) such as Poyang lake boundary, DEM, NDVI image, water level imagery, water level annual change and daily change of Poyang Lake, and climate variables (precipitation, temperature and climate-vegetation system). The boundary data of the Poyang lake basin is obtained from the open data of China fundamental geographic information data. The Poyang lake boundary data is obtained by digitizing the map in a printed form. The river net data, or the main tributaries of Poyang lake basin, is provided by the Helmholtz Centre for Environmental Research. All these three data layers are in vector form. To generate the Poyang lake basin DEM, combination is made for the multi-blocks of Space Shuttle Radar Topography Mission (SRTM) data (<https://earthexplorer.usgs.gov/>) processed by HASM (Yue 2011), and then extraction of Poyang lake basin DEM is made employing the abovementioned boundary. The DEM is in a raster format and the spatial resolution is 90 meters by 90 meters. The land use map in a resolution of 30 meters by 30 metres for realistic environmental display is downloaded from

Google earth. Since the study area is relatively large, the imagery comprises many blocks, which are downloaded and geo-referenced block by block employing ElshayalSmart Web onLine Software (<https://elshayal-smart.en.lo4d.com/>). After downloaded, ArcGIS is employed to mosaic all these images into one whole image.

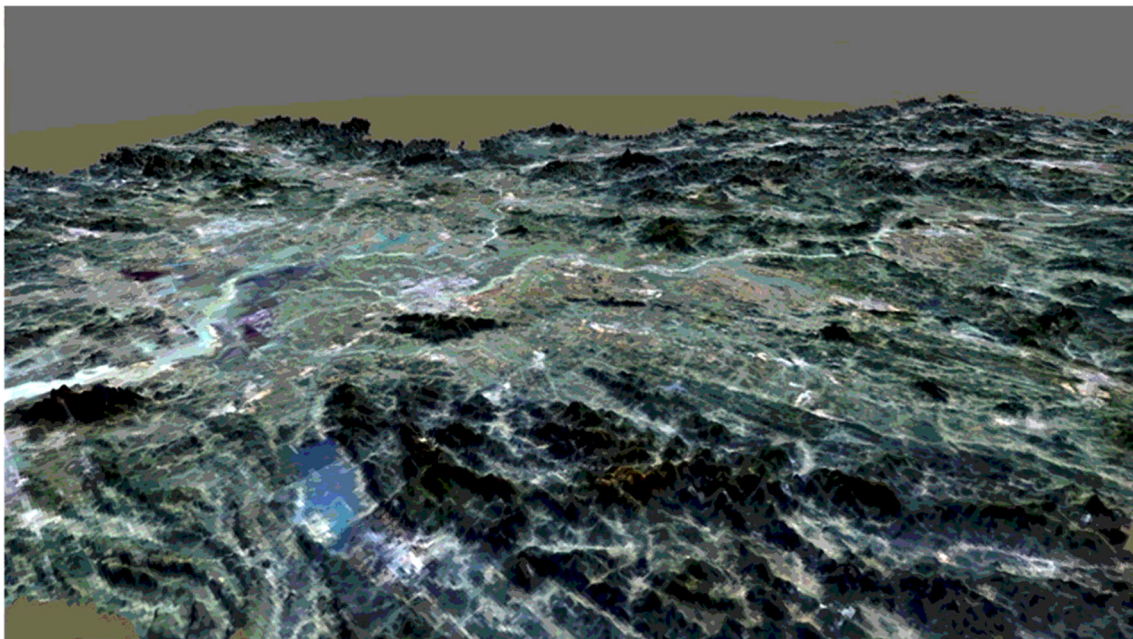
The spatial-temporal level change of Poyang lake is derived from MODIS remote sensing data (Chen et al. 2014). These images represent the yearly variation from year 2000 to 2012. The data set for the daily change of Poyang lake level is generated by the formerly mentioned hydrological model, which is a rather large file of the size about 4 Gb size (Du et al. 2018). It is split into a 183 raster files, and then geo-referenced.

With regards to the climate data sets, such as precipitation, temperature and climate-vegetation system, as aforementioned, HASM method is employed to generate them from the sampled points.

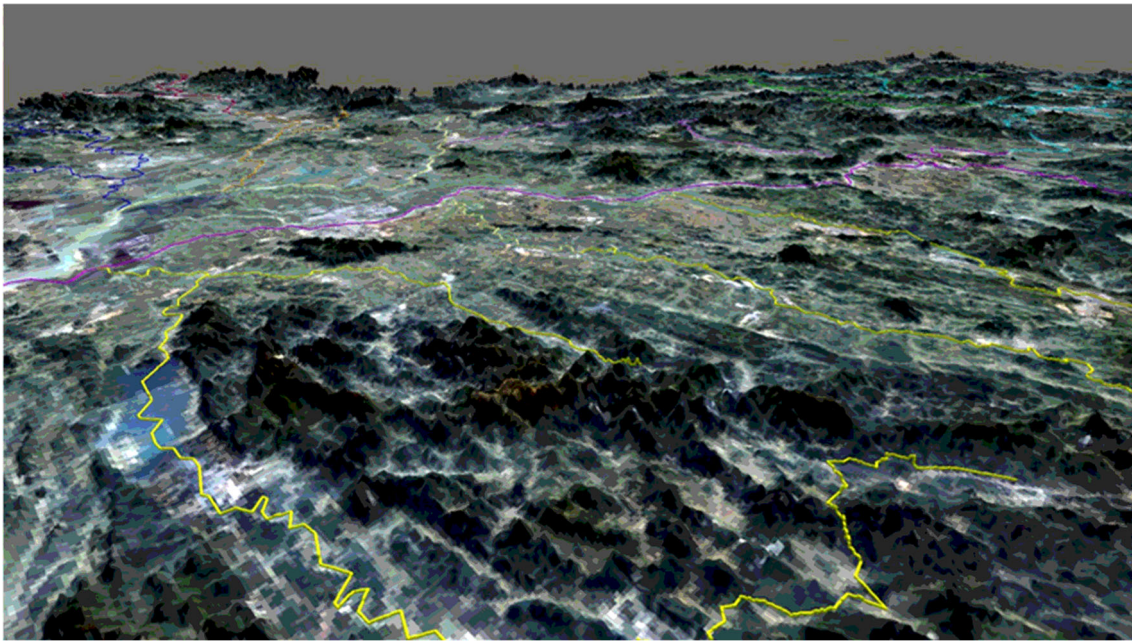
The coordinate system of all the geographic datasets above is converted to the uniform coordinate system. The geographic coordinate system adopted is GCS\_WGS\_1984 and the projected coordinate system is WGS\_1984\_Plate\_Carree. Additionally, the raster files such as time series raster files should be converted and exported as colourful images with a carefully designed colour scheme for the applications textures.

### Data layers integration

**Terrain layer integration** The terrain layer is the fundamental layer for an environmental information system (EIS). The terrain layer is composited by the DEM, mesh and the texture.



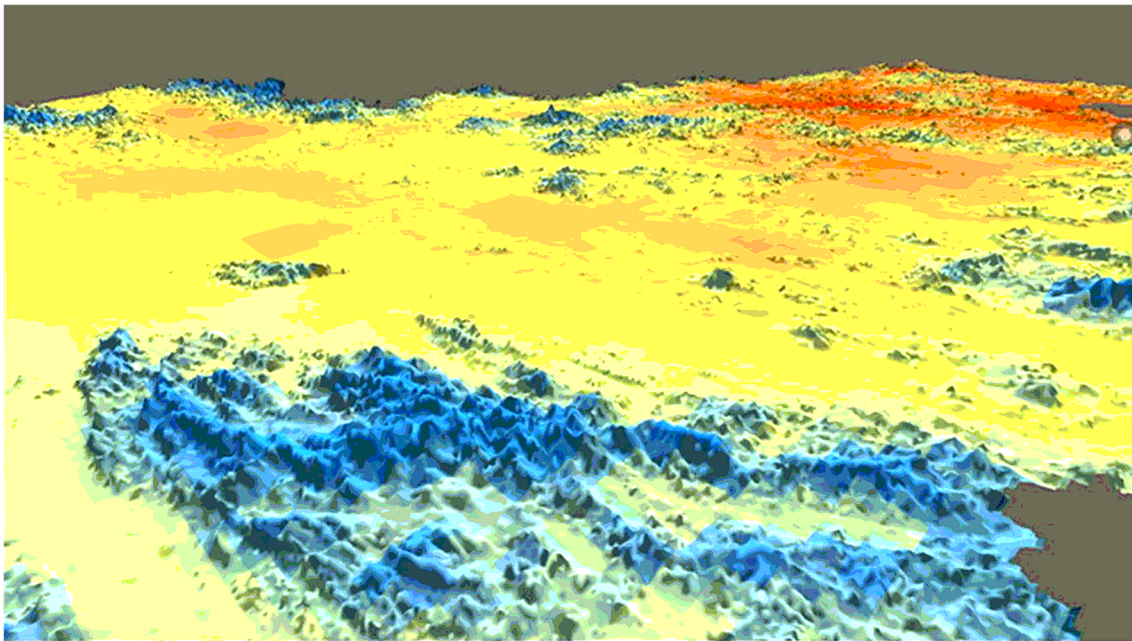
**Fig. 2** Visual integration of the terrain layer of Poyang lake basin, which shows a part of north Poyang lake basin with the Poyang Lake area covered



**Fig. 3** The river networks (polylines in different color represent different rivers) visualization, overlaid on the terrain

The integration of the terrain layer is conducted in three steps. The first step is accomplished in OpenGeoSys Data Explorer, in which the boundary data in shape file format is imported and a corresponding mesh based on the boundary is also

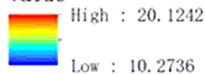
generated. Since the mesh is a flat mesh without specific elevation, it is mapped by the prepared DEM. Then a mesh with the real-world elevation is generated, which would be textured with the high-resolution imagery downloaded from Google



#### Temperature

tem2004.tif

Value



**Fig. 4** Temperature data visual integration (the area in blue denotes an area with a low temperature while the area in red denotes an area with a high temperature)



earth. The purpose of texturing the mesh is to record the coordinate of the texture in the mesh file. The mesh is then exported in a .vtp format file for later use.

In the second step, the .vtp file for the remote sensing image textured terrain is imported into ParaView for further processing for a satisfying view. After processed, the correctly textured 3D terrain landscape, is exported as Unity supported .fbx files from ParaView, which is equipped with a plugin named vtktofbx. The plugin is developed by the Helmholtz Centre for Environmental Research in Germany and can be downloaded from (<https://github.com/ufz-vislab/VtkFbxConverter>).

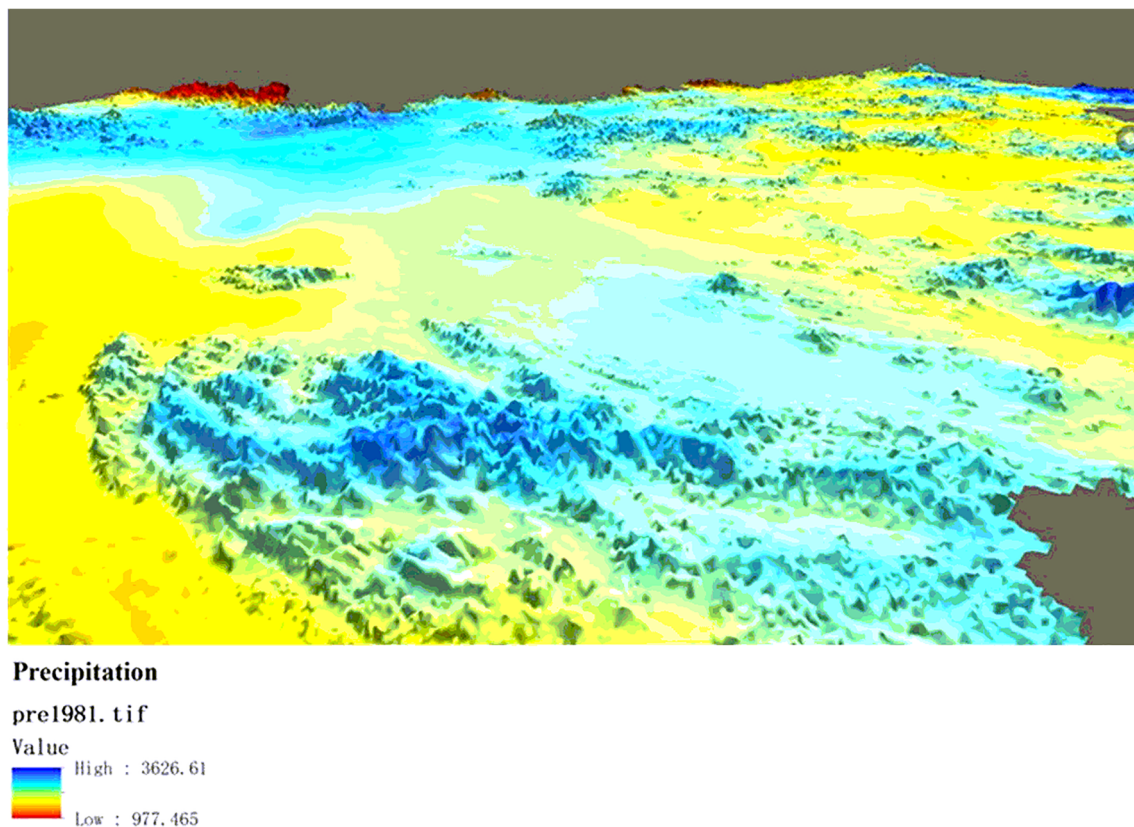
Finally, the .fbx is imported into unity and exhibited as the immersive view sight. The viewpoints or path is set to allow users a fly-through. Additionally, some texture problems or rendering options are set or fixed to obtain an ideal photo-realistic scene. The final terrain layer is showed in Fig. 2.

**River network data layer and other vector layers** River network data is added as another layer. Although the process is similar as the terrain layer integration, this data is in a vector form and requires no mesh generation. It is imported into OpenGeoSys Data Explorer from a format of shapefile. Then it was mapped with the integrated terrain. After the color configuration and the shape parameters setting for display

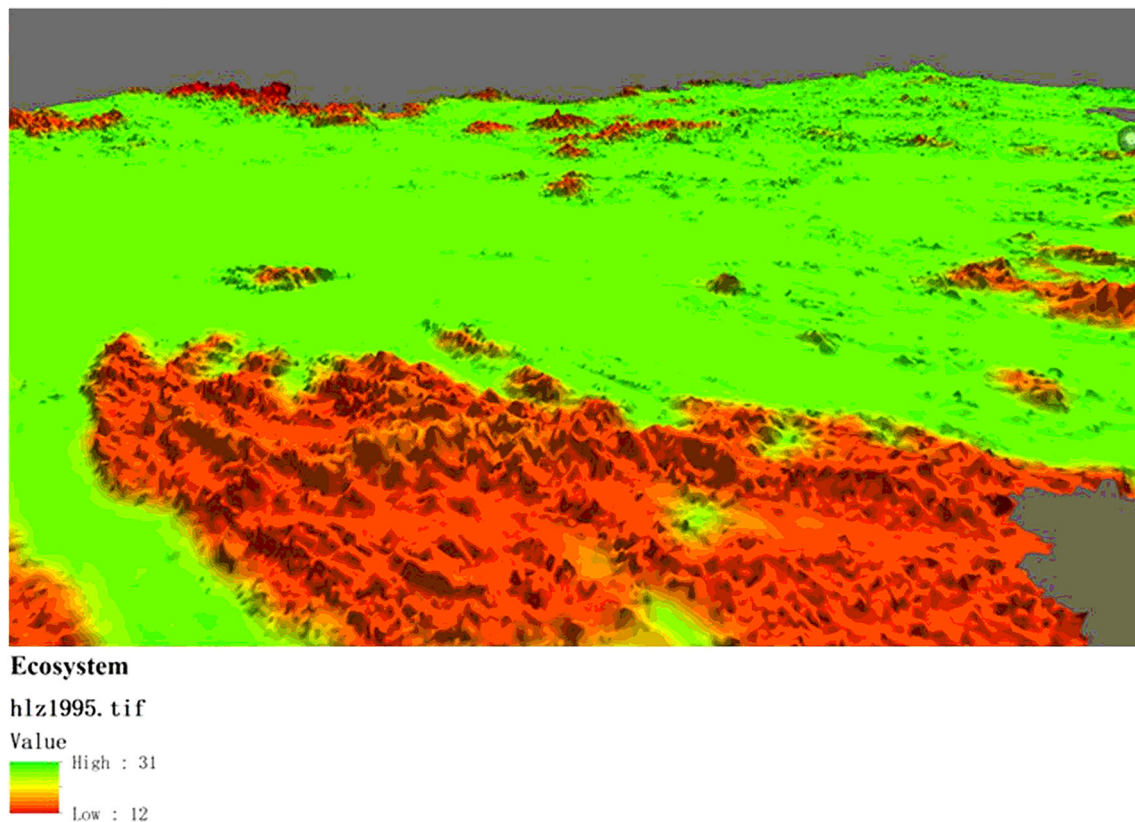
(Fig. 3), it is also exported as a .vtp file, which would be imported into unity and converted into a .fbx file. Finally, the .fbx file is imported into unity, integrated in the system, and overlaid on the terrain. Following the same routine, other vector layers can be integrated.

**Climate data surface layers integration** Climate data surfaces include temperature, precipitation and climate-vegetation surfaces, which were in a raster form. The most crucial step for these surfaces integration is to prepare them as high quality geo-referenced images. A high-resolution and a suitable color scheme are essential for visualization of these layers. Therefore, these gridded surfaces are coloured by applying the symbology tool in ArcGIS software and exported as colour textures for display. Once the textures get ready, the surfaces can be integrated, and the difference between these surfaces integration is that the surfaces should be textured with the corresponding climate surfaces. The integrated visualizations are shown in Figs. 4, 5 and 6, respectively.

**Time-series data layers integration** The hydrological data layers only cover Poyang lake area. As formerly mentioned, the time-series hydrological simulation results are prepared as



**Fig. 5** Precipitation data visual integration (the area in blue denotes an area with a high precipitation while the area in red or yellow denotes an area with a low one)



**Fig. 6** Climate-vegetation system data visual integration (the area in red denotes an area with a high climate-vegetation while the area in green denotes an area with a high climate-vegetation)

raster data format, which are generated by sampling the raster with a given time interval. Although one single file can be integrated in the similar way as terrain and climate surfaces were integrated, the time-series data should be integrated with a different approach due to the changing texture with the time. Additionally, the mesh for the Poyang Lake is in a higher resolution than the whole Poyang lake basin, due to its important role and much smaller area. Just as aforementioned, Poyang lake border, and DEM covering Poyang lake area, are required to generate the mesh. After the mesh is generated, an animation is formulated by texturing the mesh with the prepared time series raster in the different time. The annual change of Poyang Lake water level is showed in Fig. 7, which exhibits an obviously variation during the corresponding years. Additionally, daily change of Poyang Lake water level is showed in Fig. 8, which clearly indicates a significant variation during a year.

Other available time series data sets can be integrated visually according to the steps adopted on the current data sets.

### System interaction

The functions facilitating the data exploration for the user is mandatory so that the system can be well utilized. For this purpose, some necessary establishments such as the viewpoints

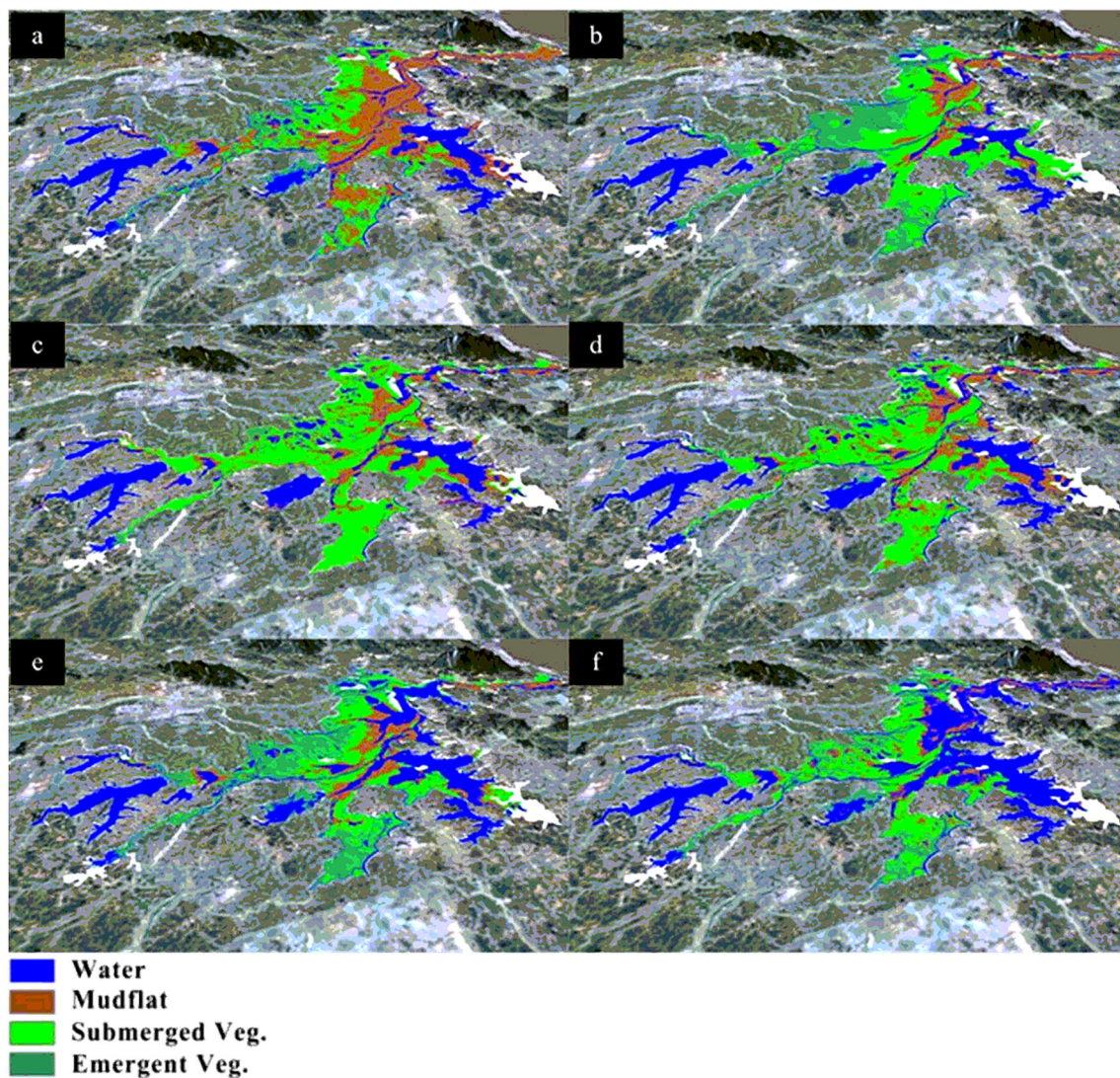
and data layers selection should be made. After the generation of the 3D visualized environment information system, the necessary viewpoints are added so that the viewers can observe the scene at certain points, and the time series data or simulation results are properly established so that they can be animated to display the visible dynamical procedures. Additionally, the visibility of every layer was established so that the users can choose which layer to explore. The interactive functionalities attribute to the aforementioned MiddleVR (Kuntz 2015), which provides the relevant interaction of the EIS so that the users can observe the scenes according to their will.

Four aspects of interactive functionalities are implemented, including viewpoints change, object selection to visualize, animations to present the time series data, and options for the basic establishment of the system (Figs. 9 and 10). Figure 9 shows current visual data layers while Fig. 10 shows the viewpoints established. Additionally, there is another menu to control the animation of the time series data.

### Platform applications

According to our framework, with all the data layers available integrated, the environmental information system or platform can be formulated. Multiple presentations using the developed





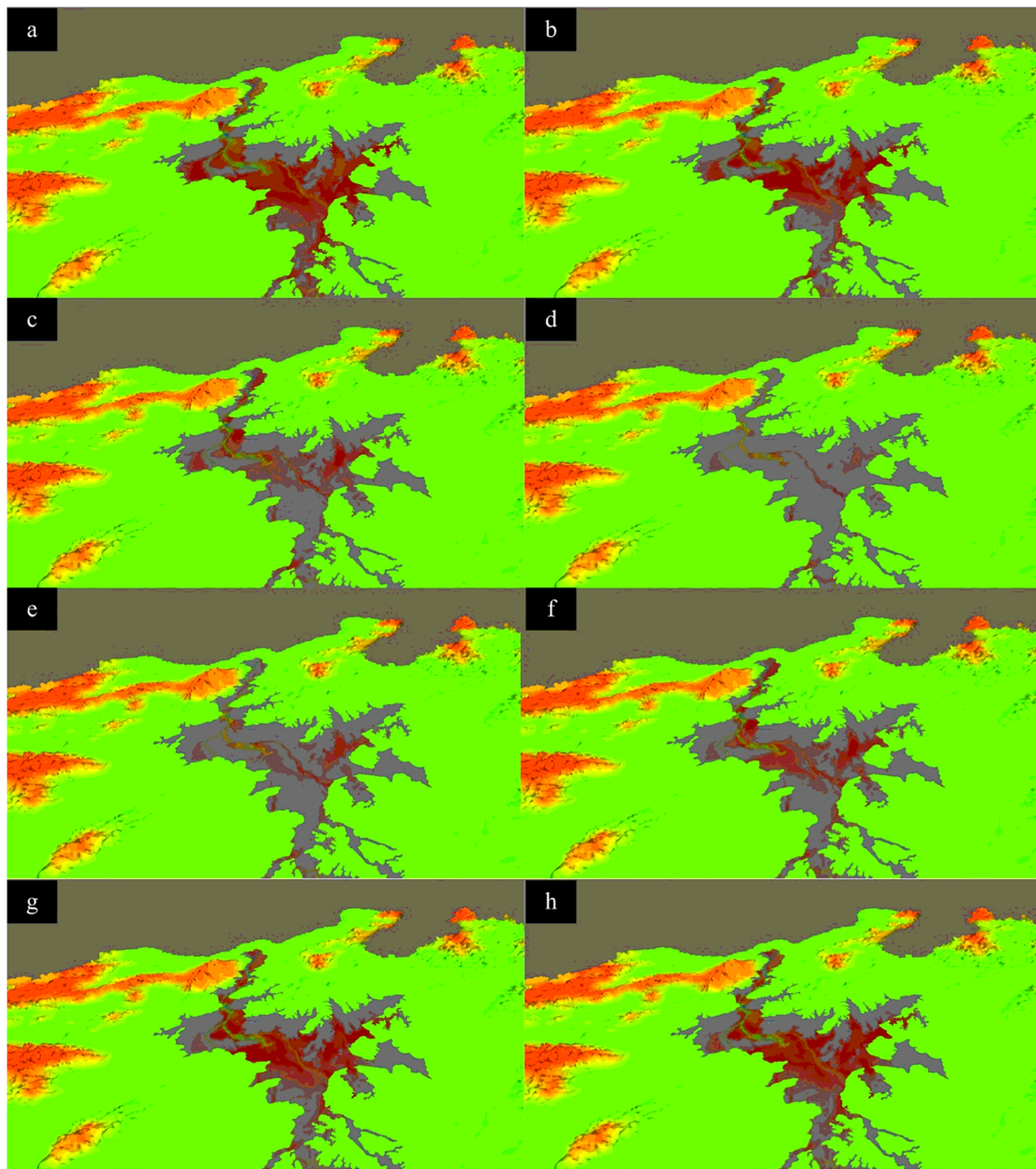
**Fig. 7** Time-series water level annual change of Poyang Lake. Six years are showed, and a, b, c, d, e, and f, represent 2000–2005, respectively

environmental information system of Poyang Lake basin were delivered. Scientists from various research domains and stakeholders involved in Poyang Lake basin have participated and rather positive responses have been received. According to the responses, at least three primary goals can be achieved. Firstly, the interrelation of the environmental data sets can be clearly discriminated and understood. Secondly, the communication between different people from different domains with different background is greatly promoted. Finally, the decision makers praised its role on decision supporting for the vivid, understandable and immersive information and environment.

## Discussions

A framework for environmental data visualization is developed. An EIS platform was developed for Poyang lake

basin within the proposed framework. This framework facilitates the visualization of the heterogeneous environmental data for the research area. With the platform, the users can concentrate on the environmental problems themselves, since the environmental data can be explored, compared, analysed, communicated and discussed in a vivid and understandable way. For instance, the correlations between the heterogeneous data sets is crucial for relevant researches as mentioned by Helbig et al. (2014). This can be clearly observed in this system (Figs. 3, 4, 5 and 6). Furthermore, it received the positive comments and response from the stakeholders due to the contribution on supporting collaborations between scientists from divergence research fields and environmental decision makers. Additionally, the framework or system is open and highly scalable, where the heterogeneous data sets that will be available in the future can be integrated with little efforts and without negative impacts on present



**Fig. 8** Time-series water level daily change of Poyang Lake. Eight days are showed, and a, b, c, d, e, f, g and h, represent orderly sampled days in year 2015–2016, respectively

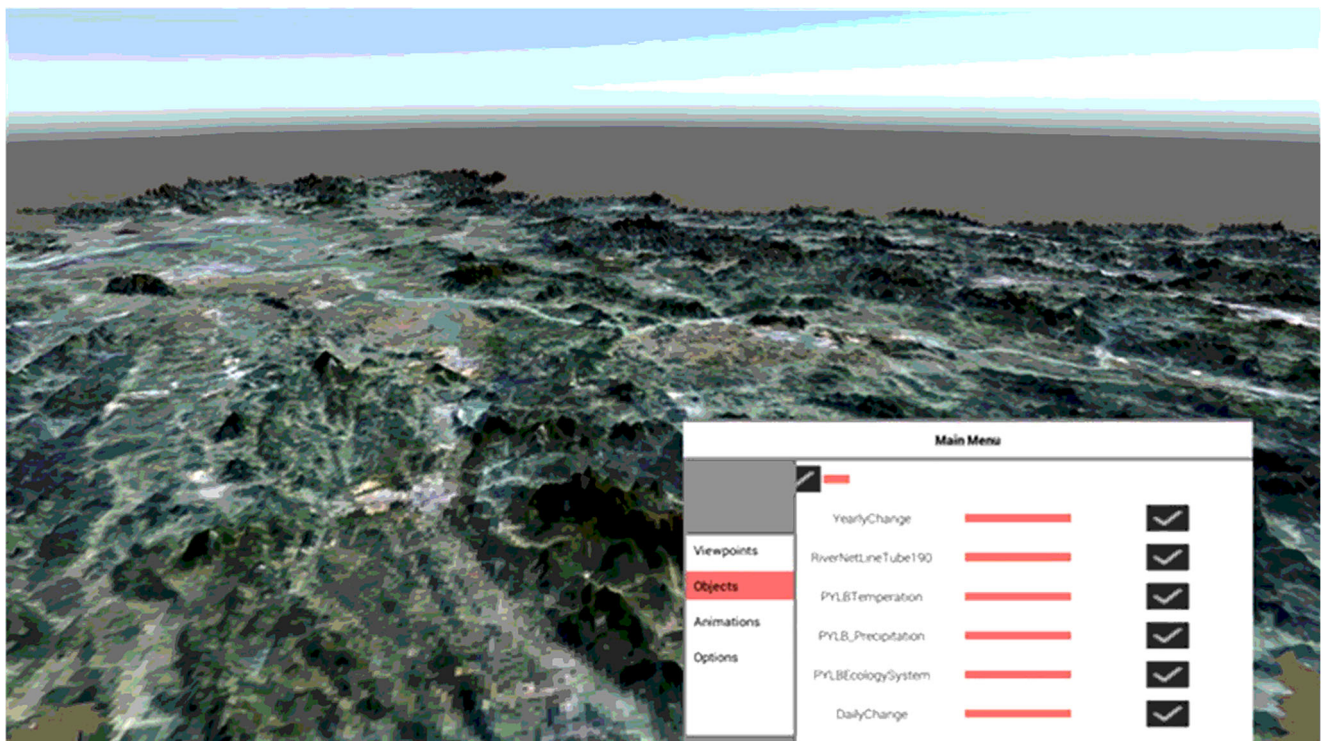
system, which is verified by the data layers integration process. This is very important for an EIS due to the continuously increasing environmental data sets.

Just as pointed out in (Chen et al. 2015), visualizing time series data is important for an EIS system, due to its evaluation and prediction functionalities. The time series simulation results generated by hydrological and land use change models can be integrated in the system, which can provide intuitive visualization for users.

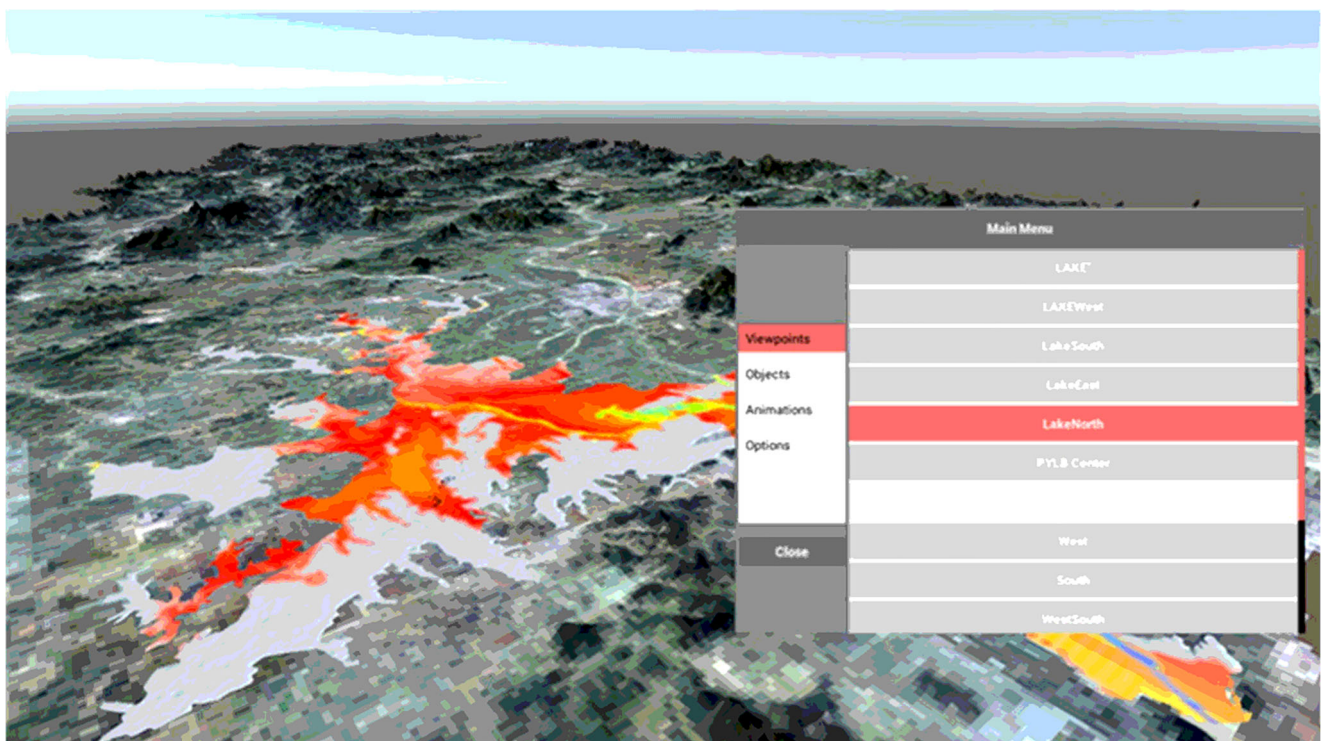
Despite the merits mentioned above, the system needs further improvement on two issues. Firstly, due to the

limitation of the unity game engine, the visualization of large scale data with very high resolution integration is not well supported, that is why the large scale data is split into multiple blocks. Trade-off has to be found between the data scale and the resolution. Further efforts should be made to overcome this problem. Secondly, since multiple software tools involved in the workflow, increasing the involvement of the system construction engineers and relevant professionals, this framework is lack of feasibility and automation, which should be further improved.





**Fig. 9** The interactive interface for visual layers establishment on the system. It is included in the main menu window for providing the interactive functions



**Fig. 10** The interactive interface for viewpoints change on the system, where multiple viewpoints are listed for selection in the main menu window



## Conclusion

In this manuscript, a framework to visualize key environmental data layers in three-dimensional space is proposed. The Poyang lake basin Environmental information System is developed as a case study of the framework, which provides an intuitive, high immersive and understandable way for environmental stakeholders. It is confirmed that with the 3D virtual environment, the efficiency of data exploration, collaborations and communications between users with different background can be promoted, thereby enabling informed decisions.

This framework and system is highly scalable, which enables various heterogeneous geo-referenced datasets integration and addresses the heterogeneous environmental data integration problem. To integrate as much environmental data as possible so that relevant decisions can be made based on sound and sufficient environmental information is the future focus. Furthermore, further improvements for automating the environmental data visual integration will be made in the future work.

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